Sections: B,C & D

Linear Algebra (MT-1004)

Homework#31

Instructor: Muhammad Ali

Ridge Regression-Case of Multicollinearity

Question 1

A real estate company wants to predict the price of houses (in thousands) based on three factors: the size of the house (in square feet), the number of bedrooms and the number of bathrooms. The dataset is as follows:

Size (Square Feet)	Bedrooms	Bathrooms	Price (Y in thousands)
2000	3	2	400
1500	2	1	250
3000	4	3	600
1800	3	2	350
2500	4	2	500

- Perform a ridge regression model on the provided dataset with $\alpha = 0.1$
- Report the coefficients b, w_1, w_2 and w_3
- Predict the price of a house when the size is 2300 square feet, the number of bedrooms is 3 and the number of bathrooms is 2